

COINPICKS · RESEARCH & EDUCATION

Bitcoin Technical Analysis Quant Pillar Report

The Tape Itself — Measured, Scored, and Mapped to the Pillar Standard

*Machine-computed from 16 years of daily price data (5,871 days, 2010–2026) through a 7-indicator scoring stack, halving-cycle analogs, and a power-law valuation model. Every number is reproducible from the scripts in the BTC-TA folder. **These numbers are computed, not judged, so they carry no draft/committed status of their own — they enter the master blend directly. The rest of today's qualitative score set was reviewed and stamped committed by Alex (see the other pillar reports); this pillar's own output is unaffected by that review.***

COINPICKS QUANT PILLAR REPORT · THE QUANTITATIVE SLEEVE OF MY THESIS

Data through August 16, 2026 (BTC \$62,937.03) · Pipeline re-run & verified August 16, 2026 · Not Financial Advice

The Scores — August 16 Model Output

Computed August 16, 2026 (previous run August 14; no cycle ran Aug 15) — same standard as every qualitative pillar. (August 14 output was 42/36/45/68/71; today's mapping nudges the 1-month window 42→41 — pure mapping noise on today's price move, the tape composites themselves held their agreement counts.) All four qualitative pillars held flat today (see the master report); this quant pillar is computed fresh every run regardless and carries no draft/committed status of its own.

WINDOW	MODEL P	QUALITY	IMPACT	SIGNAL	CALL
1 month	41	0.75	8	-1.08	Mild Bear — broken tape, repair signal #1 (200-week MA) still underwater; the 1mo cycle-analog median softened further
3 months	36	0.70	8	-1.57	Mild Bear — the final-flush window, 0-of-3 analog
6 months	45	0.55	8	-0.44	Neutral (slight bear lean)
1 year	68	0.60	8	+1.73	Bull — the cycle turn
3 years	71	0.50	8	+1.68	Bull — value + cycle

Impact: 8 (the tape aggregates every driver, but adds no causal force of its own) · **Formula:** Signal = ((P - 50) ÷ 50) × Quality × Impact · **Honesty cap:** P = 50 + 25 × (window score ÷ 100) — pure TA never claims odds more than 25 points from a coin flip, because its measured out-of-sample edge is modest. Math machine-verified.

THE THREE INPUTS, IN PLAIN ENGLISH

- P** — the odds the tape is *good for Bitcoin* by then. Near-term low (weekly composite -29.0/100, 5-of-7 indicators still bearish — price has now slipped fractionally below the 200-week average); long-term high (at this exact point in prior cycles, the next 12 months were positive 3-for-3, and price sits 54% below its power-law fair value).
- Quality** — how trustworthy the evidence is. Highest near-term (841 weekly bars, unanimous indicators), lowest at 3 years (only 3 completed cycles to learn from, and this cycle is already breaking the old pattern).
- Impact** — 8 out of 10, now the uniform value shared by every pillar; the TA pillar carries an equal 20% weight in the master, same as each qualitative pillar. Price is the thing itself — but a mirror, not an engine.

The Claim

The thesis, and where the numbers differ from the mood

- The crowd:** -50% from the all-time high, a lower low on June 30, a weekly close under the 200-week average as recently as June 28 — "the cycle is dead, get out." (Price closed at \$62,937.03 (+7.5% off the June 30 low), up 0.2% vs. Aug 14 even as every qualitative pillar held flat — the tape moving on its own schedule while the qualitative news cycle stayed quiet is itself information — it isn't the thing driving crypto right now.)
- The model:** the short-term tape is still broken (P 41/36 for 1–3 months — this is not yet a dip-buying report), and the 6-month window is still a near coin flip with a slight bear lean (P 45). But the same data says P 68 at one year: every prior cycle at this exact month looked just as dead and was higher 12 months later, price is deep below fair value, and the froth that precedes real crashes is absent.
- The edge:** the crowd reads one clock (the tape). The model reads three — tape (bearish), cycle clock (bottoming window: prior cycles bottomed at months 26–30; we're at month 27.8), and valuation (53% below the power-law fair

line, under the -1σ "cheap" band). Two of the three clocks are already on the bull's side; the tape is the last one, and it always turns latest.

The Facts — The Tape

Computed August 16, 2026 from Bitstamp daily data (BTC \$62,937.03, daily close as of August 16). Current weekly/monthly/quarterly bars are still forming — scores use them as-is. Quantified: on completed bars only, the weekly composite is -25.0 , monthly -32.6 , quarterly $+7.7$ — unchanged from Aug 14, since only the still-forming current bar moved.

- **Weekly composite $-29.0/100$ (Slight Bearish), 5-of-7 indicators agree.** Price is below the 20-week (\$69,066, -8.9%) and 50-week (\$82,468, -23.7%) averages and now sits **fractionally below the 200-week (\$63,996, -1.7%)** on a weekly basis; RSI 38.9; MACD line below zero ($-4,903$) with histogram positive ($+419$) — the momentum-repair signal is weaker but not dead; %B 0.27.
- **The first repair signal remains underwater.** Price closed Aug 16 at \$62,937.03 vs \$63,996 weekly-basis MA (-1.7%), a marginally smaller gap than Aug 14's -1.9% — the actual weekly bar (which is what confirms or breaks this signal) isn't closed until Sunday. Still unconfirmed until a weekly close clears the 20-week average ($\sim \$69.1K$) — the real confirmation level.
- **Monthly composite $-41.9/100$ (Slight Bearish), 7-of-7 agree** — the full monthly indicator stack remains unanimously bearish. Below the 20-month average (\$87,439, -28.0%), still above the 50-month (\$60,739, $+3.6\%$); RSI 43.2; monthly MACD line $-1,044$, histogram $-5,772$ — deeply negative; %B 0.16 — near the lower band.
- **Quarterly composite $+8.3/100$ (Neutral), agreement 1-of-7** — the long tape still holds its bull structure (above the 20-quarter average, \$57,369, $+9.7\%$) but with falling momentum (RSI 53.1, MACD histogram $-1,428$) and heavy quarterly OBV distribution. The weakest, noisiest layer of the stack.
- **Levels that matter:** support \$59,930, then \$57,735. Resistance \$74,434, then \$81,685 (2x touch), then \$89,164; the falling weekly support trendline sits at \$46,575, while the falling resistance trendline sits at \$78,442.
- ★ **June 30 remains the cycle low: \$58,526 close** — the flush tagged the Fib golden pocket (\$54,257–\$57,802 = 61.8–65% retrace of the \$15,479 → \$126,272 run) and has held a bounce to **\$62,937.03 (+7.5% off the low)**. **\$58,526 is still the line in the sand.**



Weekly tape, scored $-29.0/100$ with 5 of 7 indicators bearish: below the 20-, 50-, and (fractionally) 200-week averages, broken trendline, MACD regime negative. Machine-annotated from the same data that feeds the scores.

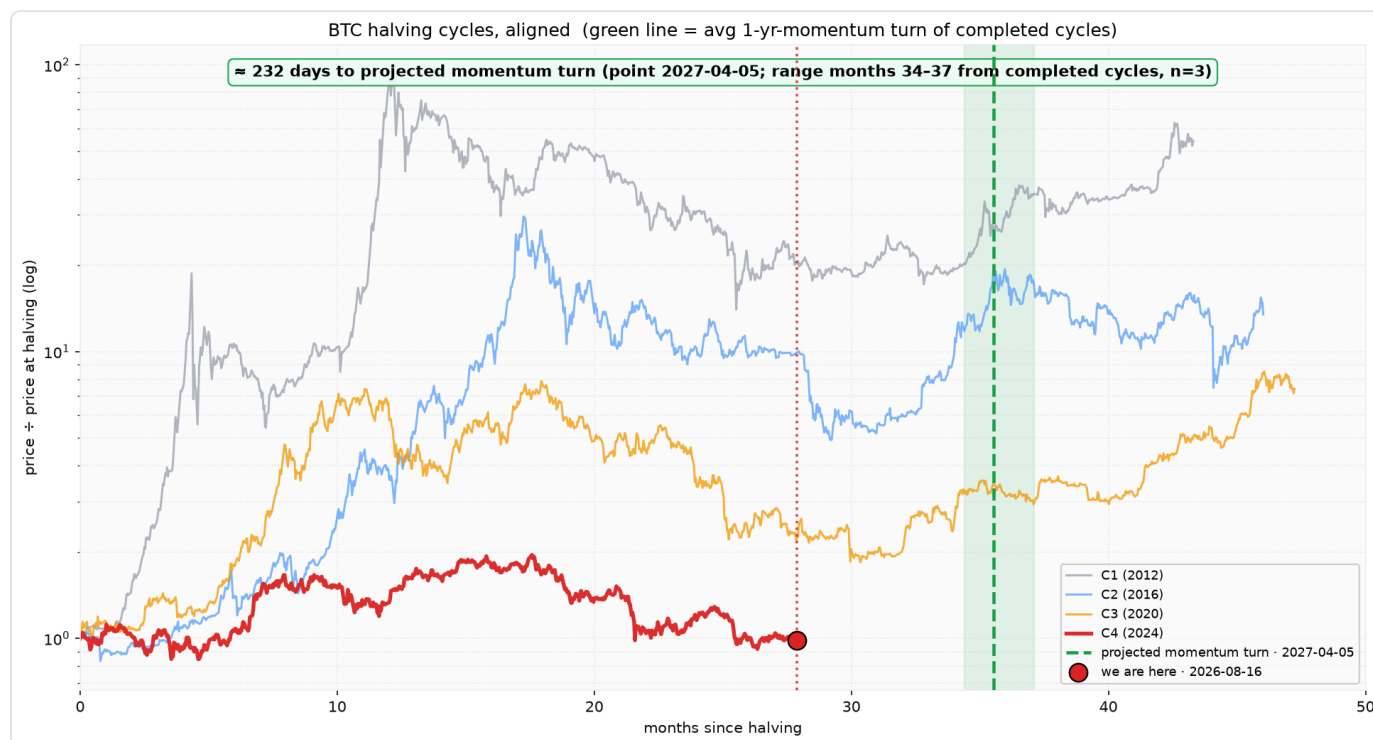
The Facts — The Cycle Clock & The Value Floor

The two slower clocks. Same data, longer lenses.

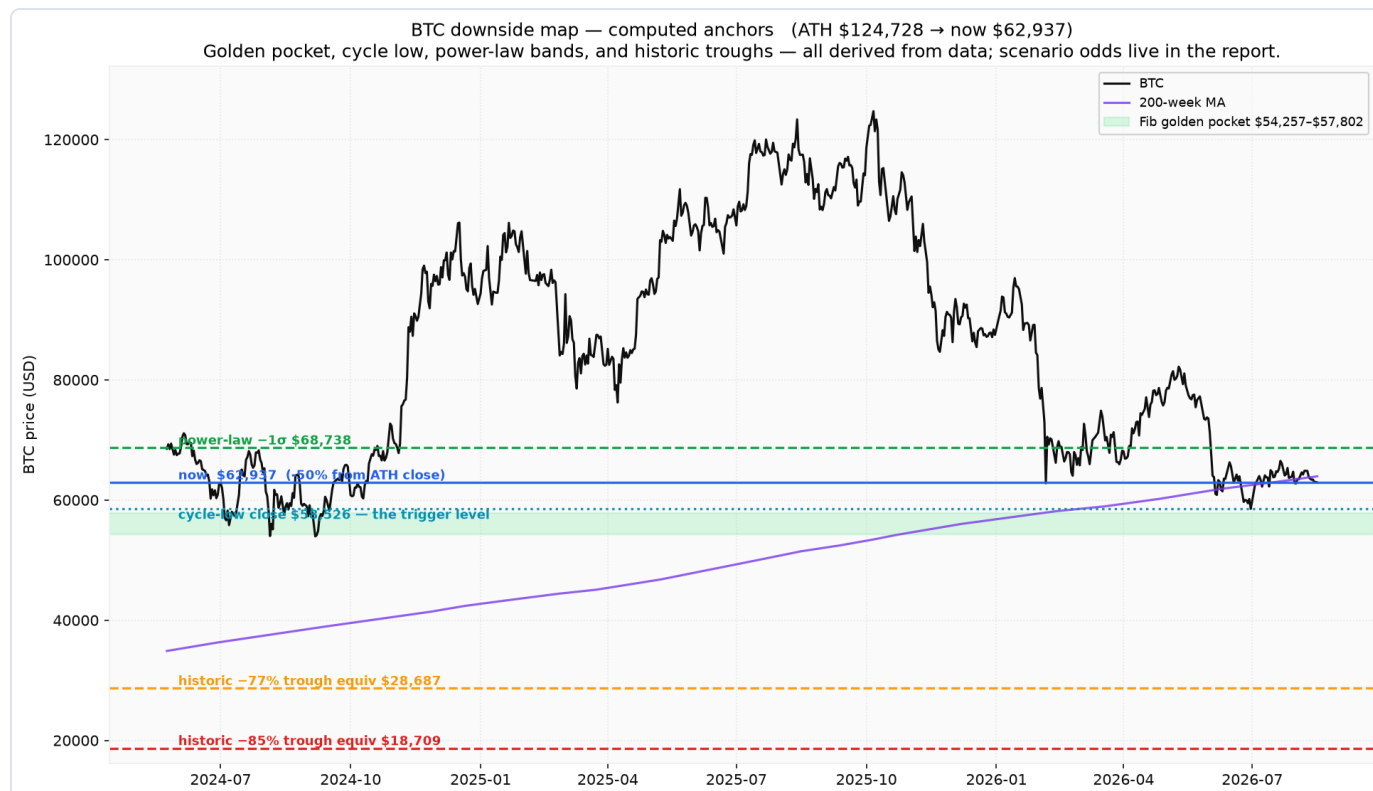
- The halving clock says "bottoming window."** The three completed cycles topped 12–18 months after their halving (avg 15.8) and bottomed at months 26–30 (avg 28.4). This cycle topped at month 18 (Oct 6, 2025 — \$124,728) and printed its low-so-far at month 26 (June 30). **We are at month 27.9; the average-bottom date still projects to ~Aug 31, 2026.**
- ★ The cycle analogs are the spine of the long-window scores — and drifted again this cycle, a real fragility worth flagging.** Standing at this exact cycle-month in C1/C2/C3: next 1 month positive **1-of-3** (median -11.5% , softer than Aug 14's -6.8%) · next 3 months positive **0-of-3** (median -12.7%) · next 6 months positive **1-of-3** (median -5.1%) · next 12 months positive **3-of-3** (median $+45.6\%$) · next 36 months positive **3-of-3** (median $+859.8\%$). Bearish quarter, still-soft half-year, bullish year — the data's own words. *Fragility disclosure, live example: every one of these medians moved a few points purely on day-count alignment as the cycle-month clock advanced one more day — exactly the kind of swing this report has warned $n=3$ analogs are prone to. Treat the analog levels as soft, the direction as the signal. And know what the 36-month number is: from month ~ 27 that window reaches month ~ 63 — it contains the NEXT halving's rally.*
- 1-year momentum turned positive at months 34–37 in every completed cycle (avg 35.5) → projects to ~April 2027** (80% window Jan–Jul 2027) for this cycle.
- The power-law floor:** a log-log fit over the full 16-year series (R^2 0.96) puts fair value at **\$137,708**. Fit-window honesty: starting the fit in 2011, 2012, 2013, or 2015 instead moves fair value to \$127,653 / \$132,645 / \$116,251 / \$143,161 respectively (price ~ 44 -54% below fair across that whole range), and R^2 this high is near-automatic for this kind of regression — the level is soft, but the **"below the -1σ cheap band"** verdict holds across every

window tested (-1σ \$68,738; -2σ \$34,311). Every halving-era cycle trough was -77% to -85% from ATH (\$18.7K–\$28.7K equivalent today) — the pre-halving 2011 bear went $\sim -93\%$ ($\sim$ \$8.7K equivalent), excluded from the range but in the data; this cycle is -54% so far.

- **No froth in the system:** the altcoin-froth index reads $z = -0.87$ — the opposite of the euphoric signature that preceded prior deep crashes. Validated on 16 years of data: high-froth days led to -39% average 12-month drawdowns vs -23% for low-froth days (gap significant, 95% CI excludes zero). *Data lags ~ 6 weeks (through May 23) — used as a tilt, not a score input.*
- ★ **What we rejected:** a 5-feature logistic crash model was tested leave-one-cycle-out and **failed** (AUC 0.44 vs 0.58 for froth alone; 0.70 = useful). It is not in this report's numbers. The scores rest only on what survived out-of-sample testing.



All four halving cycles aligned (log scale, price ÷ price-at-halving). Triangles mark where 1-year momentum turned positive in each prior cycle — months 34–37, projecting to $\sim$ April 2027 for this one. The red dot is today: month 27.8, the historical bottoming window.



The downside map, updated after the June 30 flush: the projected \$54K–\$61K bottom zone was hit (\$58,526 close / \$57,735 intraday, inside the golden pocket's doorstep) and bounced. Odds re-weighted accordingly: ~40% the low is in, ~35% one more flush to \$54–\$58K, ~17% deeper bear, ~8% tail.

The Prize

Illustrative only — not price targets, not advice

From \$62,937.03: the 20-week average (~\$69.1K, trend-repair confirmation) is **+9.7%**; a retest of the \$124,728 ATH is **+98.2%**; the power-law fair line is **+118.8%**. The median 12-month analog from this cycle-month is **+45.6%** (~\$91.6K by mid-2027); the median 36-month analog is +859.8% — **discount that one hard**, it leans on regimes (2017 mania, post-COVID ZIRP) that don't repeat on demand.

The Prize doesn't feed the math — it tells you what the lean is worth acting on. Here it says: the payoff for surviving a bearish quarter is a full cycle turn.

The Other Side

The strongest honest case the model is wrong

- **n = 3.** Every long-window claim rests on three completed cycles. Three coin flips landing heads is not a law of nature.
- **This cycle already broke the pattern.** A -53% drawdown at month 26 vs -77/-85% in prior cycles could mean dampened cycles (bullish read) — or it could mean *the bottom isn't in*: a historic-depth trough from this ATH is \$18.7K–\$28.7K, far below every level in this report.
- **The ETF era changed the machine.** Prior cycles had no spot-ETF flow channel, no CME basis trade at this scale, no corporate treasuries. Analogs may simply not transfer.
- **The power law is a curve fit, not a promise.** R^2 0.96 in-sample says nothing about a structural break — a fit like this "said cheap" through most of 2022's decline too. It bounds scenarios; it doesn't time anything.

- **The tape's edge is small by construction.** The honesty cap exists because measured predictive validity is modest (froth AUC 0.58). If you removed the cap, this report would be louder — and less true.

The Triggers

Price and dated events — some prove the read right, some prove it wrong

WHEN / LEVEL	EVENT	MEANS
\$58,526	Daily close below the June 30 low	Flush scenario live — golden pocket \$54.3–57.8K is the target; re-rate 1/3-month down
~\$64,000	Weekly close above the 200-week MA (\$63,996)	The daily close (Aug 16) sits marginally closer to the line (–1.7%, vs. Aug 14's –1.9%) — the actual weekly bar hasn't closed yet; Sunday's weekly close is now the real test
~\$69,100	Weekly close above the 20-week MA	Trend repair confirmed — re-rate 1/3-month up
~Aug 31, 2026	Historical average-bottom month (28.4) passes	A higher low after this date + improving monthly momentum = basing confirmed
Jan–Jul 2027	Projected 1-yr momentum turn window (point: ~Apr 2027)	Arriving without new lows = cycle thesis on track; re-rate 1-year up
\$50,000	Weekly close below	–60% from ATH; the dampened-cycle story is failing — see the flip line

The flip line: a decisive weekly close below **\$50,000** breaks the base case and the golden pocket both — treat the dampened-cycle thesis as broken and put the historic trough zone (\$18.7K–\$28.7K) back on the table. Above it, a bearish quarter is the plan, not a surprise.

My Decision

✓ **COMMITTED — AUGUST 16, 2026 · AUTO-STAMPED BY ALEX (delegated authority)**

Committed P: 41 / 36 / 45 / 68 / 71 · Quality: 0.75 / 0.70 / 0.55 / 0.60 / 0.50 · Impact: 8

This is the quantitative sleeve of the thesis — the same five windows, same formula, same table as every qualitative pillar, but computed instead of judged. The near-term read is still the honest one: the tape is broken (repair signal #1 still underwater, still awaiting Sunday's weekly confirmation) and the 3-month analog is 0-of-3 positive — the worst window in the set. The 1-month window ticked down today on pure mapping noise, not a fresh signal. The 6-month window remains a near coin flip with a slight bear lean. The long read is where the quant leans bullish: cycle clock + valuation + absent froth.

These numbers enter the master blend as committed. Override any P and re-stamp — the blend recomputes. Unlike the four qualitative pillars, this pillar carries no draft/committed status of its own tied to today's light daily touch — it is computed fresh every run and enters automatically.

What Changed Since Last Check

Pipeline re-run August 16, 2026 (previous run August 14, 2026; no cycle ran Aug 15).

- ★ **Price ticked up: \$62,802.46 (Aug 14) → \$62,937.03 (Aug 16), +0.21%.** June 30's \$58,526 remains the cycle low; price is now +7.5% off it.
- **Repair signal #1's daily-close margin narrowed slightly:** today's daily close (\$62,937.03) sits −1.7% below the 200-week average on a weekly basis (\$63,996), marginally closer than Aug 14's −1.9% — the actual weekly bar (the real test) hasn't closed yet; Sunday's close now decides it. A decisive break above the falling resistance trendline (\$78,442) remains the real repair signal #2.
- **Composites: weekly and monthly firmed slightly on the day's price move, quarterly ticked up; completed-bar figures unchanged.** Weekly −29.4 → −29.0 (agreement holding 5-of-7), monthly −44.9 → −41.9 (agreement holding 7-of-7 — full monthly stack still unanimously bearish), quarterly +8.0 → +8.3 (1-of-7, unchanged agreement) — tracking the two-day price move, not a regime change; completed-bar composites (weekly −25.0, monthly −32.6, quarterly +7.7) are identical to Aug 14.
- **Model P: 42/36/45/68/71 → 41/36/45/68/71.** The 1-month P ticked down 1 point on pure mapping noise, not a fresh signal; every other window held. The cycle-analog medians drifted a few points across the board on day-count alignment, as they do most days — flagged as the kind of n=3 fragility this report's own disclosure warns about, not treated as new information.
- **Data refreshed:** Bitstamp daily through August 16, 2026 (5,874 days on the master series); Coin Metrics altcoin set still through May 23 (community data lags ~6 weeks — froth is a tilt, not a score input).
- **Context:** all four qualitative pillars held flat today (see the master report) — this pillar's own computed output is independent of that process and enters the master blend the same way every day.

Method & Sources

Everything is reproducible — the scripts and data live in the BTC-TA folder. Run order below.

- **Price data:** Bitstamp OHLC API (daily, 2011–2026, no key) via [fetch_btc_history.py](#) · Coin Metrics community reference price (2010–2011 close-only) via [build_master_history.py](#) — 5,835 days total, each row tagged by source.
- **Indicator stack:** [indicators.py](#) — SMA 20/50/200, EMA 12/26, RSI-14 (Wilder), MACD 12/26/9, Bollinger 20/2, OBV — on weekly (W-SUN), monthly, quarterly bars. [levels.py](#) — swing-cluster S/R + least-squares trendlines.
- **Scoring:** [score.py](#) — each indicator → conviction in [−100, +100] (bounded inputs mapped linearly, unbounded squashed with tanh); composite = fixed weights (MA .20, RSI .15, MACD .15, BB .15, S/R .15, OBV .10, TL .10); agreement = indicators sharing the composite's sign.
- **Cycle & value:** [cycle_analysis.py](#) — halving-aligned tops/bottoms/momentum turns, projections from C1–C3 · [downside.py](#) / [downside_map.py](#) — power-law bands (R^2 0.96) + historical troughs + 6-method bottom-zone synthesis · [test_altcoin_froth.py](#) — froth quartiles vs forward drawdowns, block-bootstrap CI.
- **The mapping (new):** [pillar_mapping.py](#) — window scores from tape composites + cycle analogs + power-law value; $P = 50 + 25 \times (S/100)$; Quality rubric per window; writes data/pillar_scores.json for the master blend.
- **Run order:** [fetch_btc_history](#) → [build_master_history](#) → [fetch_altcoin_data](#) → [indicators](#) → [levels](#) → [score](#) → [charts](#) → [cycle_analysis](#) → [downside](#) → [downside_map](#) → [pillar_mapping](#).
- **Honesty notes:** current weekly/monthly/quarterly bars are partial as of Aug 16 (complete-bar composites: weekly −25.0 / monthly −32.6 / quarterly +7.7) · the multi-signal crash model failed out-of-sample (AUC 0.44) and is excluded · the AUC figures are the pipeline's own attestation · froth data lags ~6 weeks · n=3 on all cycle claims, and analog windows can cross into the next cycle — the 6-month analog remains fragile on pure alignment drift, printed in the Facts · the 2024 halving is dated Apr 19 (US ET; block 840,000 hit Apr 20 00:09 UTC — 0.03 cycle-month effect) · S/R levels are algorithmic swing-clusters with 1–2 touches each · the bottom-odds split (40/35/17/8) is a judgment synthesis, not a model output.